



UE24MA241B

LINEAR ALGEBRA AND ITS APPLICATIONS

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Introductory Lecture

These slides are compiled by LA TA Sidharth Rajendran

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UE24MA241B Linear Algebra: Course Content

UE24MA241B: Linear Algebra and Its Applications - (4-0-0-4-4)

Unit 1: Matrices, Gaussian Elimination and Vector Spaces

Introduction, The Geometry of Linear Equations, Gaussian Elimination, Singular Cases, Elimination Matrices, Triangular Factors -LU decomposition and Row Exchanges, Inverses and Transposes, Inverse by Gauss-Jordan method, Symmetric Matrices.

Self-Learning Component: Algebra of Matrices.

14 Hours

Unit 2: Four Fundamental Subspaces & Linear Transformations

Vector Spaces and Subspaces (definitions only), Linear Independence, Basis and Dimensions, Row reduced Echelon form, The Four Fundamental Subspaces, Rank-Nullity theorem. Linear Transformations, Algebra of Linear transformations.

Self-Learning Component: Examples of Vector Spaces and Subspaces.

14 Hours

Unit 3: Orthogonalization, Eigenvalues and Eigen vectors

Orthogonal Vectors and Subspaces, Orthogonal Bases, Cosines and Projections onto Lines, Projections and Least Squares. Orthogonalization, The Gram-Schmidt Orthogonalization process, Introduction to Eigenvalues and Eigenvectors, Properties of Eigenvalues and Eigenvectors, Cayley-Hamilton theorem (statement only), Diagonalization of a Matrix.

14 Hours

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Unit 4: Singular Value Decomposition

Quadratic Forms, Definitions of Positive definite, negative definite, positive semi-definite, negative semi-definite, indefinite forms and matrices, Tests for Positive Definiteness, Cholesky's method, Singular Values and Singular Vectors, Image Processing by Linear Algebra, Principal Component Analysis (PCA by the SVD and Covariance matrix), Minimum Principles. **14 Hours**

Text Books:

- 1: Linear Algebra and its Applications, Gilbert Strang, Thomson Brooks/ Cole, 5th Edition, Second Indian Reprint 2007.
- 2: Application : Applied projects for an introductory linear algebra class by Anna Zemlyanova.
3. Higher Engineering Mathematics, B S Grewal, Khanna Publishers, 44th Edition, 2020,.

Reference Books:

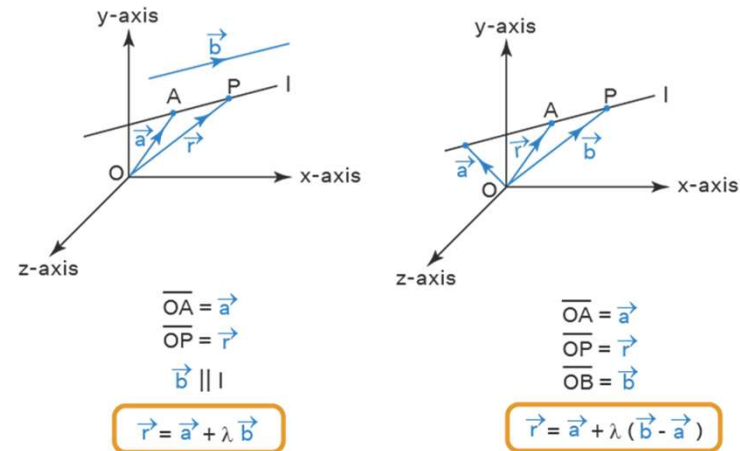
- 1: Linear Algebra and its Applications, David. C lay, Publication by Pearson Education, 5th Edition, 2015
- 2: Linear Algebra, Schaum's outlines, Seymour Lipschutz and Marc Lipson, Tata McGraw-Hill publications, 4th Edition, 2009.
- 3: Practical Linear Algebra, Gerald Farin and Dianne Hansford, CRC Press, Taylor & Francis Group, 3rd Edition.

Linear Algebra: Evaluation Policy

Pattern	ISA: 50 marks ESA: 50 marks
ISA1 & ISA2	Mode: Hybrid Each ISA will be conducted for 40 marks and scaled down to 20 marks (each). Pattern for ISA: No. of 1 mark Questions(MCQs) = 16 No. of 2 marks Questions(MCQs) = 4 No. of 4 marks Questions(Descriptive)=4 Marks Distribution and Total Marks(of each ISA): $(16 \times 1) \text{ mark} + (4 \times 2) \text{ marks} + (4 \times 4) \text{ marks} = 40 \text{ marks}$ Scaled Down marks(from each ISA) = 20 marks Total Marks(from both ISA1 and ISA2) = $20 + 20 = 40 \text{ marks}$.
JOB Problem	10 marks

Linear Algebra: A Vital Tool for Engineers

Linear algebra is foundational for engineers, enabling structural analysis, optimization, and modelling in areas like robotics, computer graphics, and signal processing.



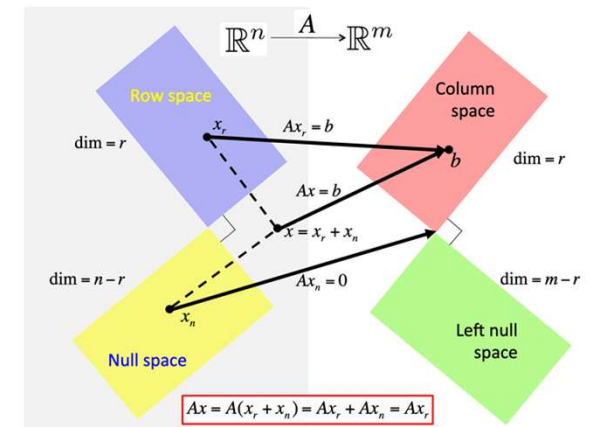
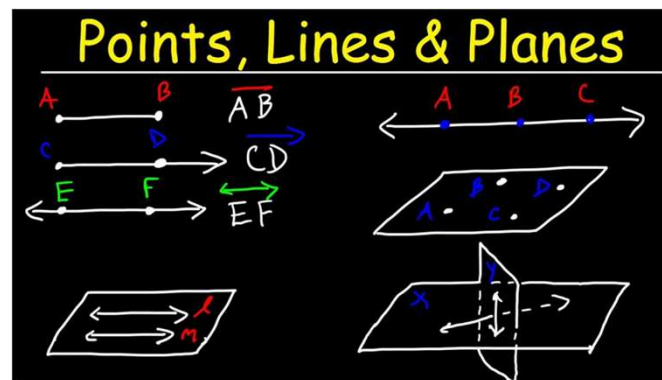
Linear algebra is essential in computer science for tasks like machine learning, graphics, cryptography, and data processing. It powers algorithms for training models, image processing, 3D transformations, and optimizing systems, making it a cornerstone for innovation in technology.

What is Linear Algebra?

It's like the ultimate cheat code for understanding the world around you.

- It's about **lines, planes, and spaces** — the building blocks of everything geometric.

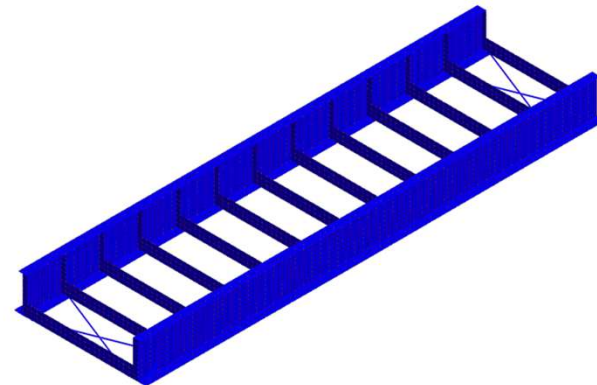
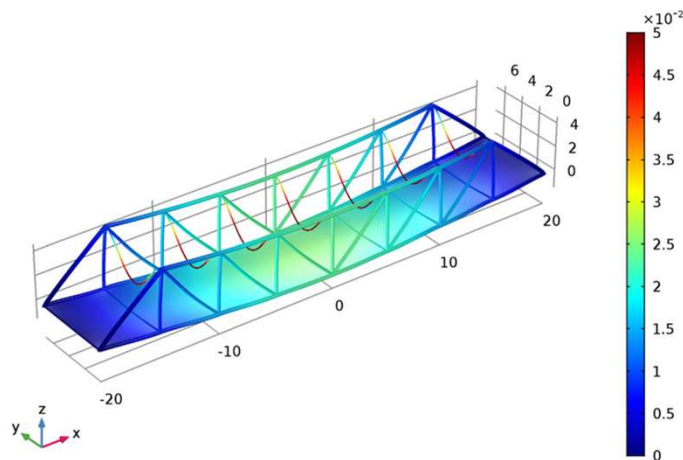
It's not just math; it's a way to **visualize and shape entire dimensions**, whether 2D, 3D, or beyond.



Applications? So Many!

- **Structural Engineering**

Think of a bridge: engineers use matrix equations to figure out how the bridge distributes weight from vehicles, ensuring that the structure stays intact under stress.



[DATA] Linear Buckling - All Bracing, Eigenvalue(Required), MODE 2 (EIGENVALUE=1.0532e+003), [UNIT] N, mm, [Output CSys] Default

Applications? So Many!

- **Electrical Engineering**

Heard of Fourier Transform?

It is essentially a **set of linear mappings** that map a time-based signal (a function of time) to a set of sinusoidal functions (functions of frequency). In engineering applications like **audio processing, telecommunications, or image processing**, matrices are used to represent these transformations and to compute them efficiently.

From designing efficient circuits to processing signals for communication systems, Linear Algebra provides the tools to understand, model, and optimize complex systems.

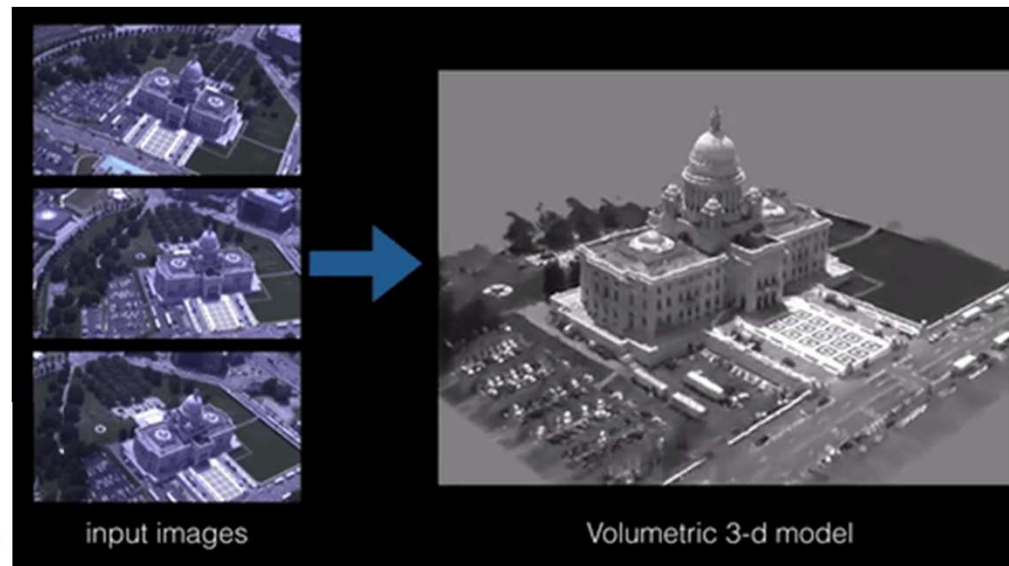
Applications? So Many!

- Computer Graphics 🎮

Transforming 3D Objects Using Rotation and Scaling Matrices:

If you want to scale a 3D model of a character to make it larger or smaller for a video game scene, Linear Algebra helps adjust its proportions by scaling its coordinates.

The image showcases how **linear algebra** transforms multiple 2D aerial photos into a single detailed 3D model of a capitol building.



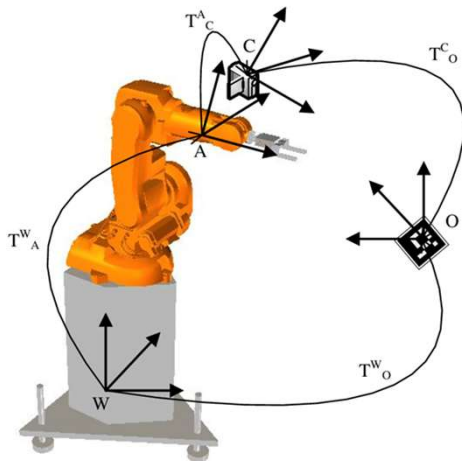
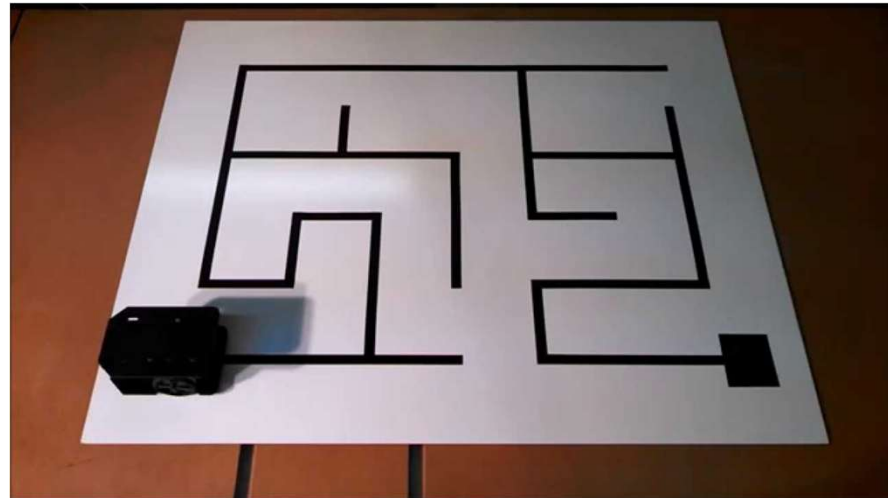
It's a powerful demonstration of using mathematical matrices and transformations to convert flat images into rich three-dimensional representations.

<https://connectjaya.com/applications-of-linear-algebra-in-data-science/>

Applications? So Many!

- Robotics 🤖

Robot in a Maze: The robot can use matrices to solve a system of equations that describe the maze's layout. By solving these equations, the robot determines its best path while avoiding walls or obstacles.



Manipulating Robot Arms using Transformation Matrices:

A robotic arm can use transformation matrices to adjust its position and orientation, calculating how far to move along each axis to pick up an object.

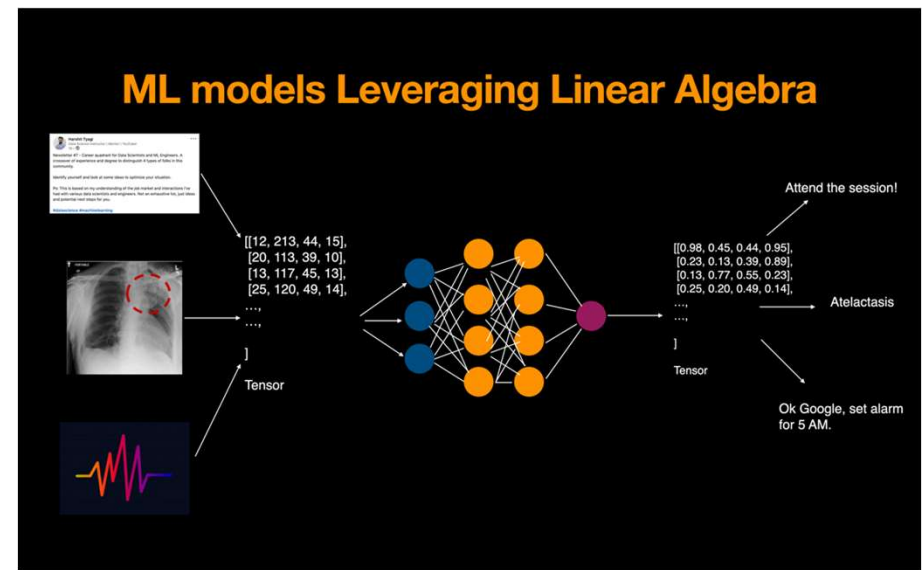
Applications? So Many!

- AI/ML

Why This Matters in AI & ML:

- Efficient Data Representation:** Linear Algebra allows us to represent data in a format that algorithms can understand and process efficiently, enabling the **training** of models and making predictions.
- Powerful Algorithms:** Many of the most powerful AI and ML algorithms (like **PCA**, **Neural Networks**, **SVM**, etc) are deeply rooted in Linear Algebra, ensuring that models are **efficient**, **accurate**, and **scalable**.

<https://www.freecodecamp.org/news/how-machine-learning-leverages-linear-algebra-to-optimize-model-training-why-you-should-learn-the-fundamentals-of-linear-algebra/>



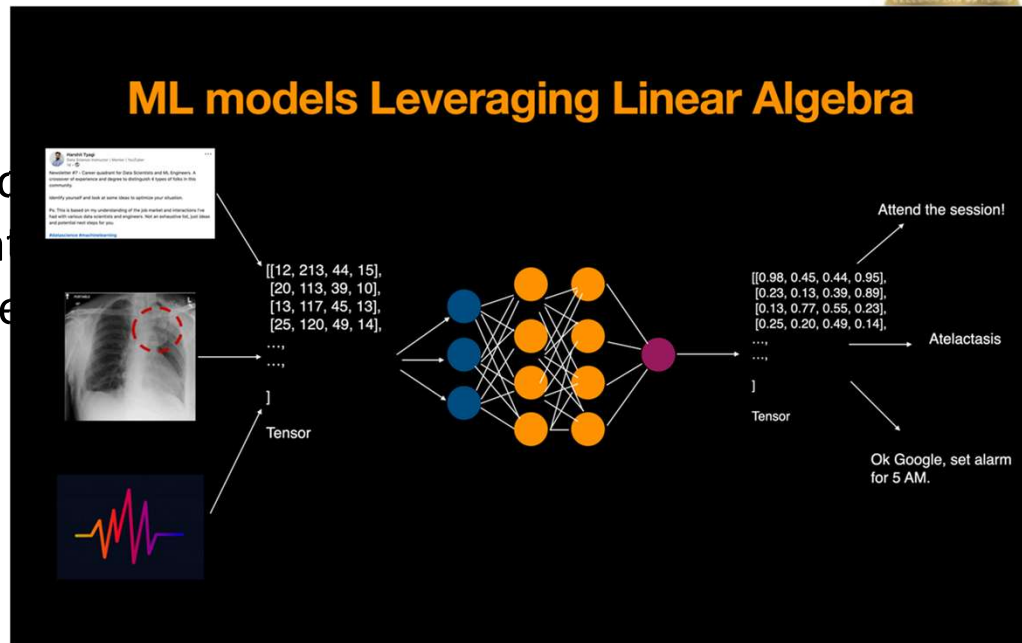
Linear Algebra is more than just a tool—it's the language through which we describe and understand data, optimize algorithms, and train models.

Applications? So Many!

- AI/ML

Why This Matters in AI & ML:

- Optimization:** Linear Algebra is used to **optimize** model parameters, ensuring that the models can learn from data and improve performance over time.



Linear Algebra is more than just a tool—it's the language through which we describe and understand data, optimize algorithms, and train models.

Applications? So Many!

- This is just the tip of the iceberg.
- In upcoming units we will look into many applications of LA pertaining to CSE.



THANK YOU
